

Pruebas con wavScalogram

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1 Windowed Scalogram Difference

1.1 Procesamiento de datos

```
getSymbols(c("SPY", "QQQ", "GLD"), from = "2021-01-01", to = "2024-12-31", src = "yahoo")
```

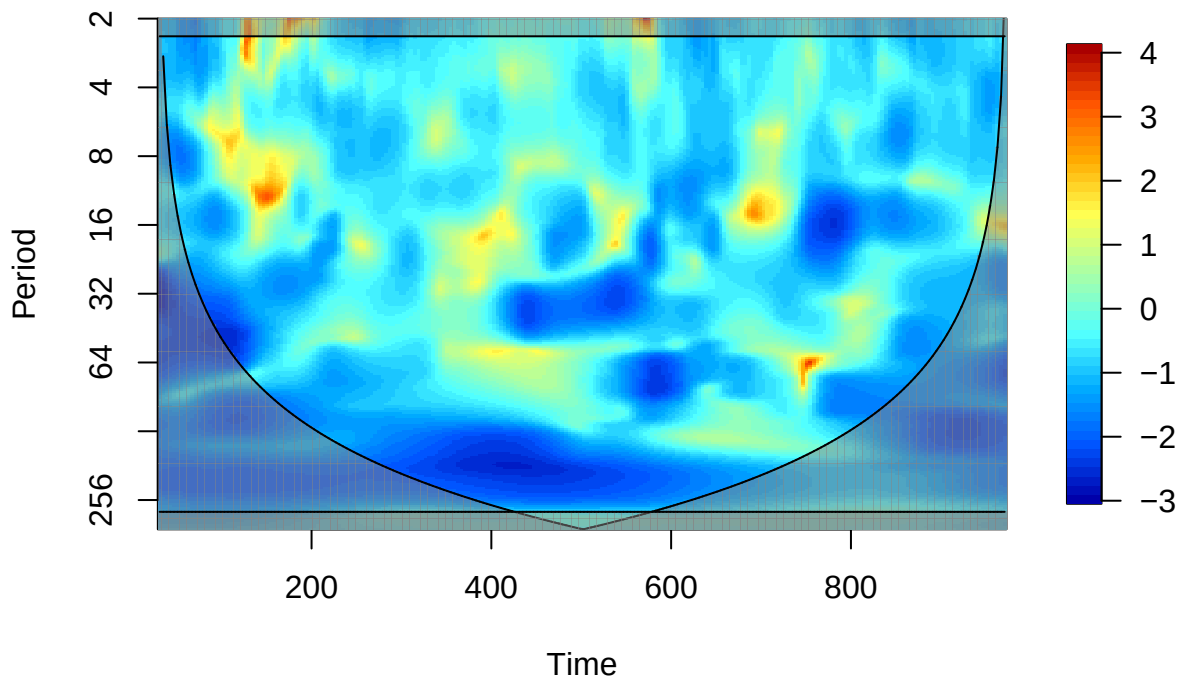
```
## [1] "SPY" "QQQ" "GLD"
```

```
# Rendimientos  
r_spy <- as.numeric(diff(log(Cl(SPY)))[-1])  
r_qqq <- as.numeric(diff(log(Cl(QQQ)))[-1])  
r_gld <- as.numeric(diff(log(Cl(GLD)))[-1])
```

1.2 Gráficos

```
wsd_corr <- wsd(r_spy,  
               r_qqq,  
               dt = 1,  
               windowrad = 30,  
               rdist = 10,  
               energy_density = TRUE,  
               makefigure = TRUE,  
               main = "SP500 - Nasdaq 100")
```

SP500 – Nasdaq 100

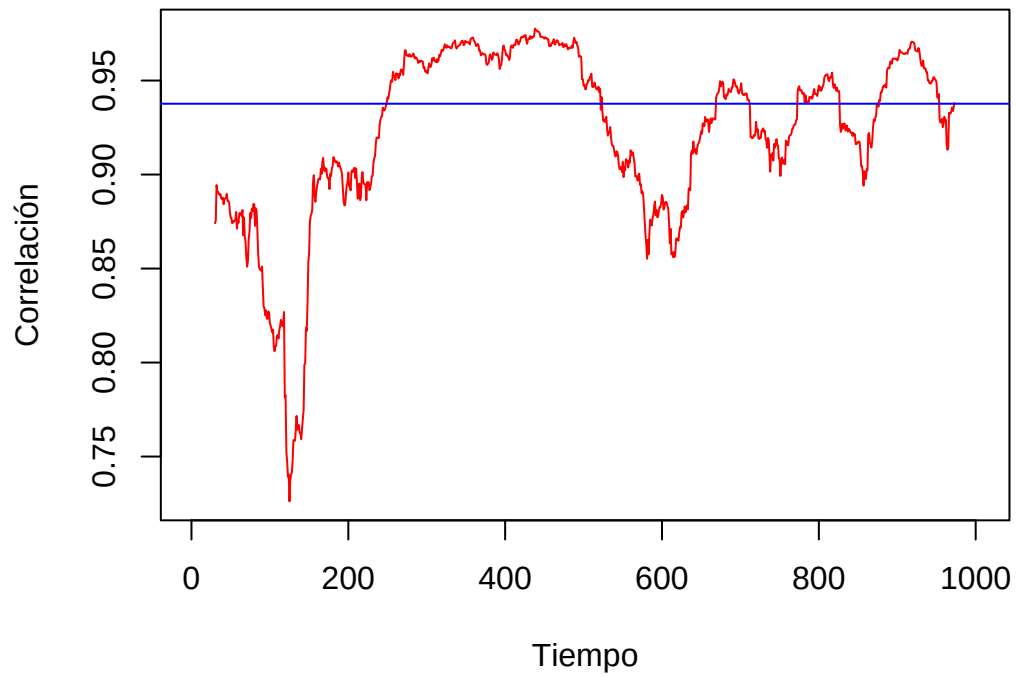


```
cor_1 <- cor(r_spy, r_qqq)

p_1 <- rollapply(cbind(r_spy, r_qqq),
               width = 60,
               function(x) cor(x[,1], x[,2]),
               by.column = FALSE, align = "center", fill = NA)

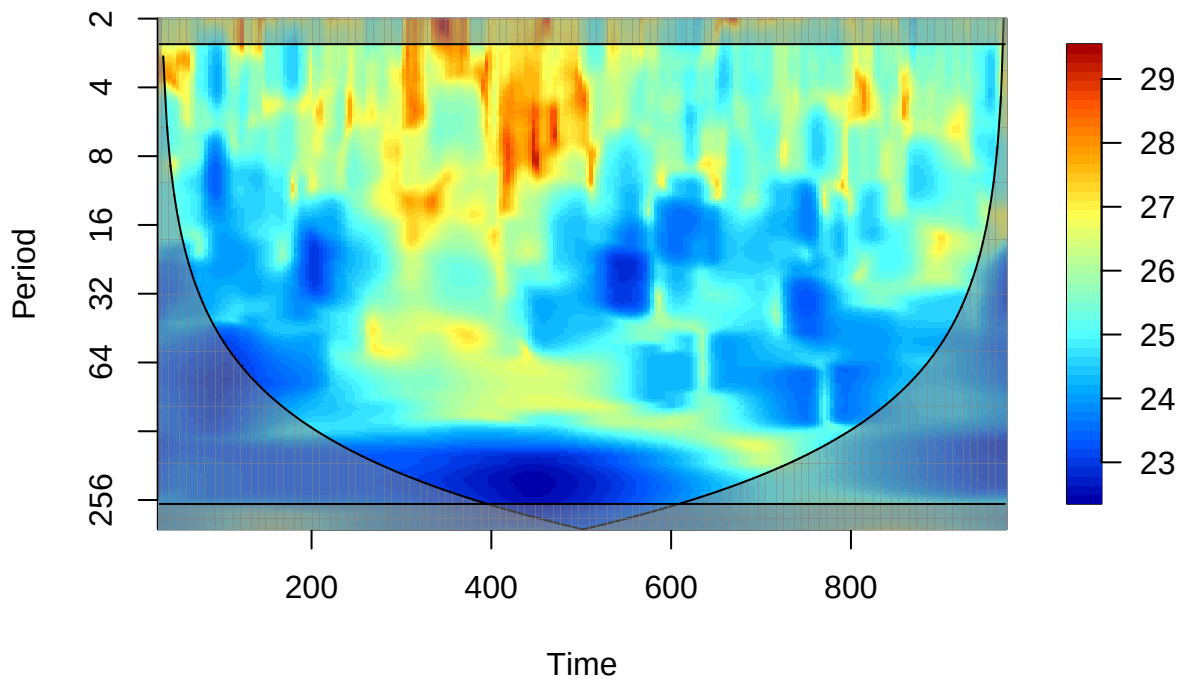
plot(p_1, type = "l", col = "red",
     main = "Pearson",
     ylab = "Correlación", xlab = "Tiempo")
abline(h = cor_1, col = "blue")
```

Pearson



```
wsd_self <- wsd(r_spy,  
  r_spy + rnorm(length(r_spy), sd = 1e-10),  
  dt = 1,  
  windowrad = 30,  
  makefigure = TRUE,  
  main = "SP500 consigo mismo")
```

SP500 consigo mismo

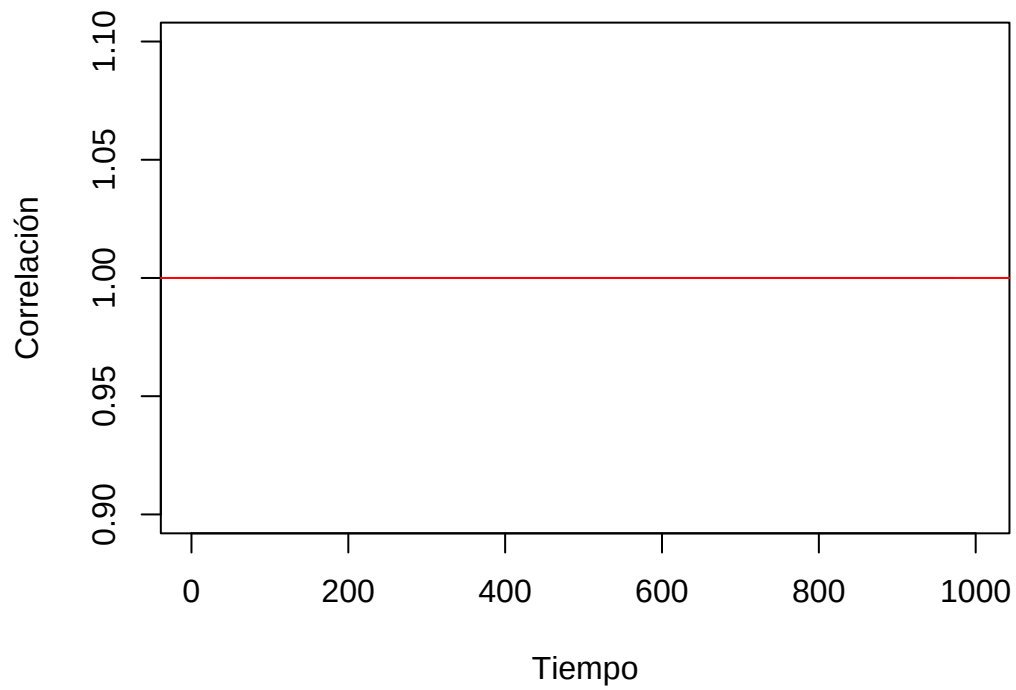


```
a <- r_spy
b <- r_spy + rnorm(length(r_spy), sd = 1e-10)

p_2 <- rollapply(cbind(a, b),
                 width = 60,
                 function(x) cor(x[,1], x[,2]),
                 by.column = FALSE, align = "center", fill = NA)

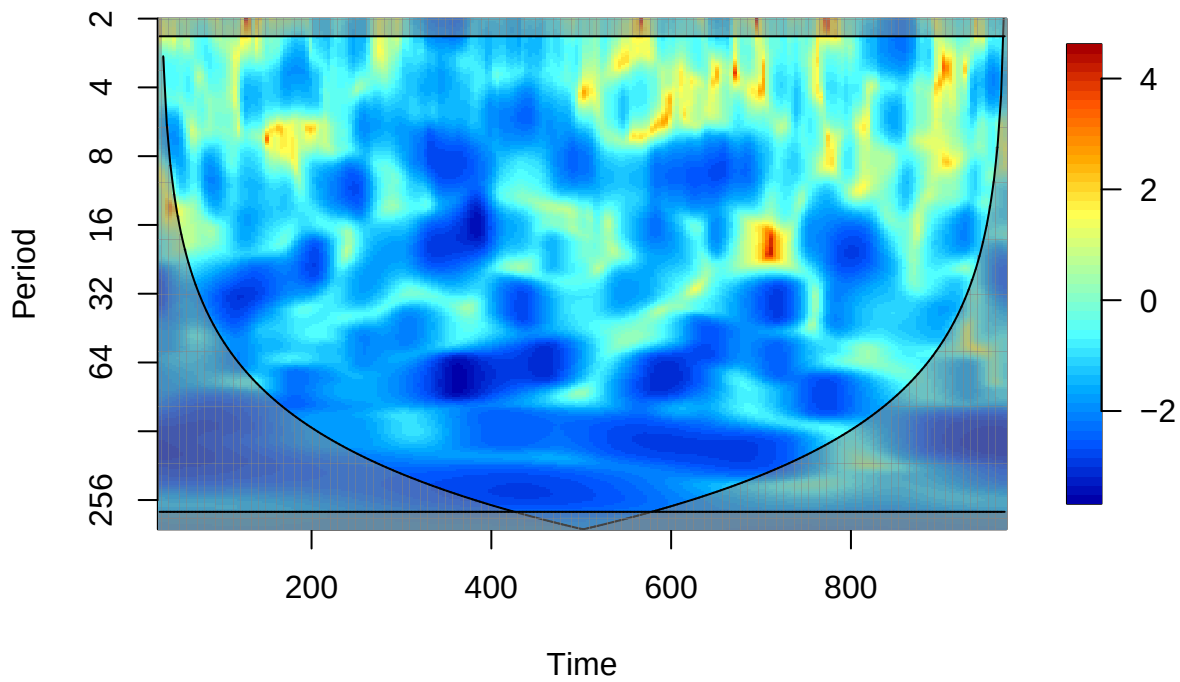
plot(p_2, type = "l", col = "blue", ylim = c(0.9, 1.1),
     main = "Pearson: SP500 - SP500",
     ylab = "Correlación", xlab = "Tiempo")
abline(h = 1, col = "red")
```

Pearson: SP500 – SP500



```
wsd_3 <- wsd(r_spy,  
             r_gld,  
             dt = 1,  
             windowrad = 30,  
             rdist = 10,  
             energy_density = TRUE,  
             makefigure = TRUE,  
             main = "SP500 - Oro")
```

SP500 – Oro

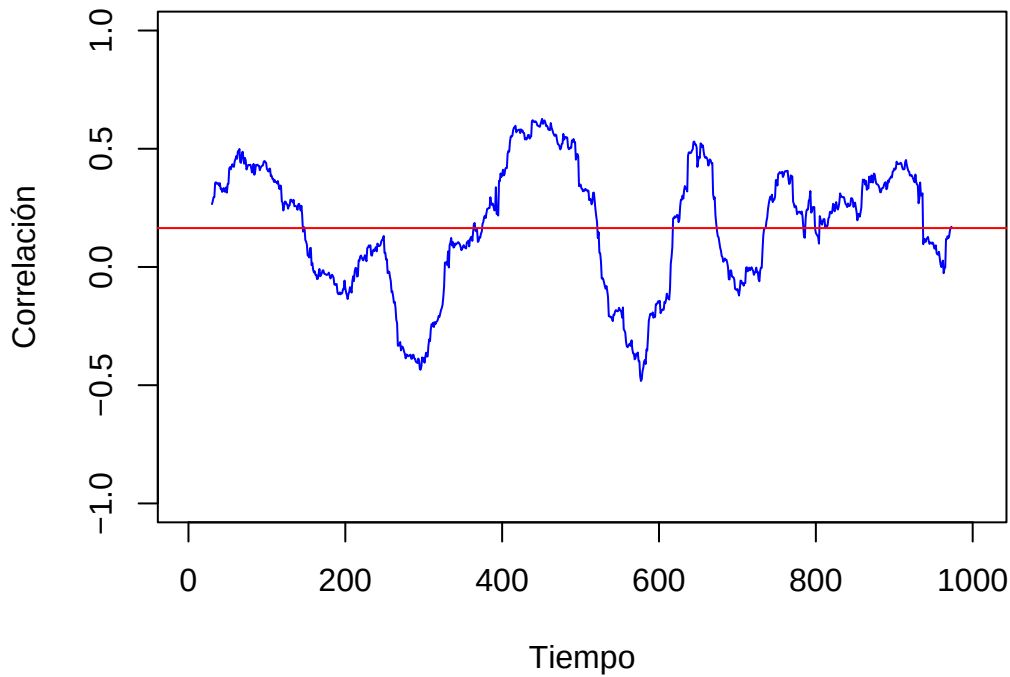


```
cor_3 <- cor(r_spy, r_gld, use = "complete.obs")

p_3 <- rollapply(cbind(r_spy, r_gld),
                width = 60,
                function(x) cor(x[,1], x[,2]),
                by.column = FALSE, align = "center", fill = NA)

plot(p_3, type = "l", col = "blue",
     main = "Pearson: SP 500 - Oro",
     ylab = "Correlación", xlab = "Tiempo", ylim = c(-1, 1))
abline(h = cor_3, col = "red")
```

Pearson: SP 500 – Oro



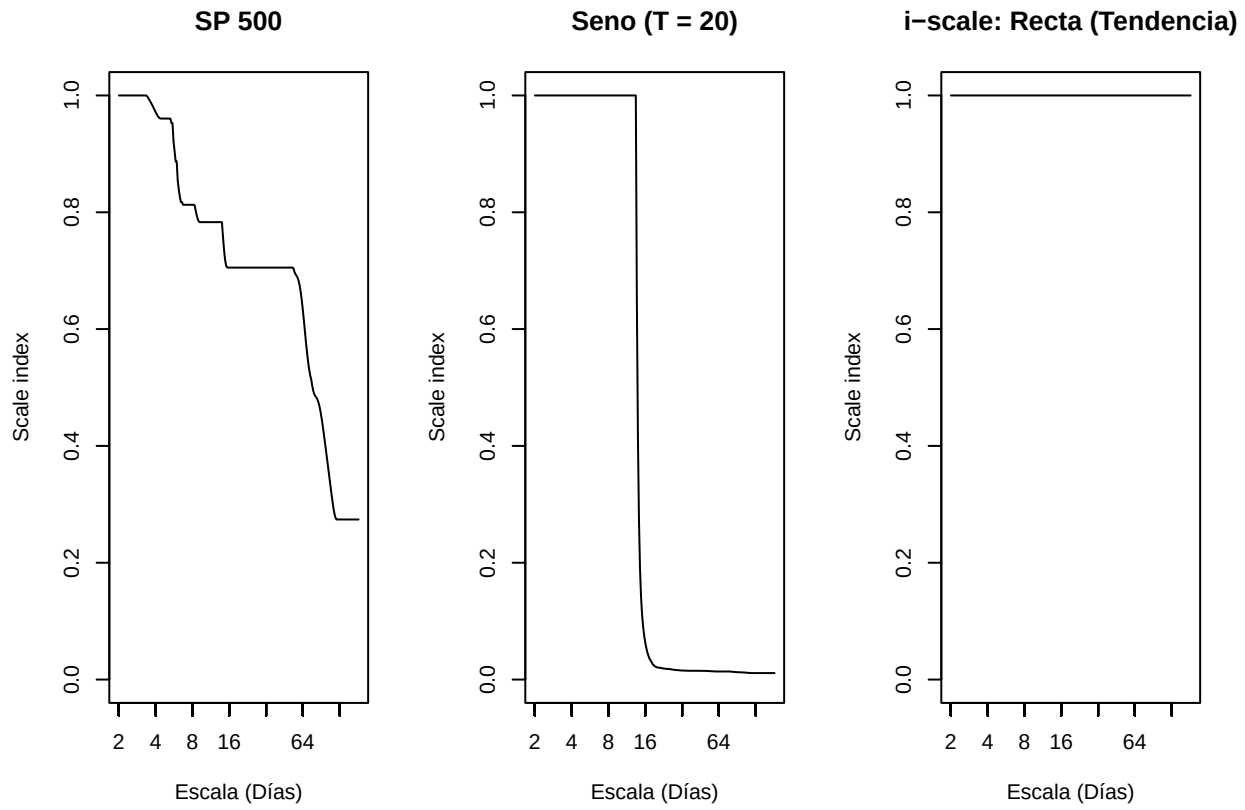
2 Índice de Escala

```
par(mfrow = c(1, 3))

tmp <- scale_index(signal = r_spy, dt = 1,
  main = "SP 500",
  xlab = "Escala (Días)")

t <- 1:length(r_spy)
sen <- sin(2 * pi * t / 20) + rnorm(length(t), sd = 0.01)
tmp <- scale_index(signal = sen, dt = 1,
  main = "Seno (T = 20)",
  xlab = "Escala (Días)")

recta <- 0.05 * t + 10
tmp <- scale_index(signal = recta, dt = 1,
  main = "i-scale: Recta (Tendencia)",
  xlab = "Escala (Días)")
```



```
par(mfrow = c(1, 1))
```